

Clothing Suite: Interactively Design Complex Garments

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Figure 1: Examples of different garments created with the Clothing Suite.

ABSTRACT

In this talk, we discuss the proprietary toolset created to move our clothing look-design process from 2D into 3D. The Clothing Suite is an easy to use and interactive toolset to create complex, art-directed garments constructed from woven curves. The Clothing Suite allows artists to build and decorate textiles in real time through a three-dimensional sculptural approach while accommodating a highly deformed animation style. This allows the look-dev artist to design the clothing in 3D space in a fast and highly iterative process instead of relying on refined flat art or complex geometry manually sculpted in the model.

CCS CONCEPTS

• **Computing methodologies** → **Parametric curve and surface models.**

KEYWORDS

surfacing, look-dev, garments, design, clothing

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1 INTRODUCTION

In recent years, the clothing designs in our movies have become increasingly complex, displaying a wide variety of fabrics, threadings and decorations. Our previous methods for achieving a garment's look consisted of applying texture maps with an anisotropic shader on geometry and covering it with a layer of fuzz. The results did not look sufficiently tactile and the artists had to spend significant time and effort to reproduce a three-dimensional look with two-dimensional textures and shaders. Because they were attempting to mimic the look of fabric, rather than focusing on details and decorations, it led to slow turnaround of iterations.

Therefore, we developed our Clothing Suite: a set of tools to easily create physical woven clothing with complexity that mimics reality with a stylized twist. These tools not only achieve highly sophisticated looks, but also provide the look-dev artist the ability to iterate quickly and effectively with the art department. This allows for the conceptual exploration of a variety of looks within the 3D space, instead of iterating with 2D paintings, significantly reducing the back and forth needed between the two mediums.

Research on woven clothing has had an emphasis on creating believable fabrics that have a look mimicking reality ([Hoffman et al. 2020] [Leaf et al. 2018] [Smith et al. 2018]). Starting from a technology similar to the one presented in [Smith et al. 2018], the Clothing Suite expands its capabilities to be able to generate an endless variety of fabrics, shading effects and decorations, while providing further flexibility and controls for fast iteration in support of the design process on a 3D medium. The Clothing Suite provides the artist the ability to layer, sculpt, customize depth, directionality and shape of the woven pattern. Moreover, we improved the manner the threads are bound to an animated driver mesh, which can be highly deformed in some animation styles.

So far, 11 of our artists have used it to author 45 pieces of garments across five different productions.

2 FABRICATING THE TEXTILE

2.1 Fabric Creation

To procedurally generate the fabric, the Clothing Suite generates a grid on the driver mesh in UV space. This driver mesh is the simplified model representing the garment. The grid is then transferred on the 3D surface of the driver mesh and used to instantiate unit woven patterns, which are preset curve sectionals of common textiles. Different textiles can be easily layered on top of each other or seamlessly stitched together.

2.2 Adding Design Details

Once the base fabric is generated, the artist can modify the woven surface by displacing it to accommodate other layers of details. The Clothing Suite provides tools to automatically generate a distance map between two objects (e.g. between the base fabric and stitching), which can then be used as a guide to paint a displacement or color map for added details like fabric puckering caused from stitching. On the base fabric the artist can also choose to layer different patterns on top of the initial weave, to create complexity in the design. Some of these patterns include: braids, ribbing and custom hand-drawn patterns. Another typical detail carried by all garments is stitching. The Clothing Suite provides tools to automatically extract and easily modify curves from a mesh. These curves can then be used as guides for stitches, and transformed in UV space. The Clothing Suite provides flexibility to create further designs, like embroideries, lace and any custom seam. The artist can paint a simple 2D tileable texture, and use that as a source image to automatically create 3D curves that are then placed on the model and used as guides for stitching. Finally through the Clothing Suite, the artist can automatically refine the generated textile by cleaning and fixing intersections where threads penetrate other meshes.

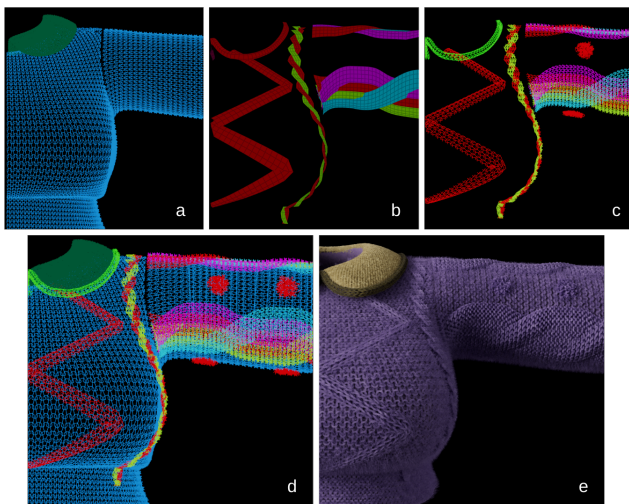


Figure 2: Assembled layers for a sweater: base weave (a), guide grid for woven braids (b), woven braids and decorations (c), assembled curves (d), rendered result (e).

2.3 Binding Textiles to Animation

Similarly to the workflow described in [Hoffman et al. 2020], we export the woven textile in rest-position, and then bind it to the base mesh through our proprietary procedural, which binds the textile and the simulated driver mesh by matching pairs of points. To better control the binding, we expanded its capability to be able to perform a matching based on automatically-generated labels. The fabric strands that compose the woven textile are defined by nurbs curves with width attribute. This width attribute can stay constant with small deformations of the base mesh. However, width should be compensated when the fabric is significantly stretched, or the space between the strands becomes too sparse and the look of the fabric is not preserved. Our proprietary procedural has the ability to deform attributes attached to the base mesh, and we use this to deform the frenet frame of each curve CV. By comparing the length of the vectors in the rest pose versus their length in the deformed pose, local stretching and compression can be measured to modulate the width attribute. This supports our animation style which features a wide variety of extreme non-uniform deformations of the characters. However, the artist can choose to apply this method based on the artistic requirements.

2.4 Shading

The Clothing Suite provides simple ways to generate attributes on threads or sets of threads, which can be single thread or sub-thread based, driven by a noise, based on the directionality of the thread, manually painted, etc. These can then be used in the shader networks to drive the CV color to provide a realistic or stylized shading effect, which is not achievable with a traditional texture application.

3 CONCLUSION

The Clothing Suite is an interactive toolset for garment designers that is intuitive, highly iterative, and powerful. The layering nature of the tools allows for designs to be as simple or complex based on art direction. The procedural nature of the suite allows for designs to be highly iterative while supporting downstream departments' workflows, revolutionizing our design process for clothing.

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